

DNA Recombination

Current 31-page faculty deck primary - official Week 4 objectives integrated - repair, linkage, transposons, and immune rearrangement connected

5 official objectives

HR vs NHEJ

HLA linkage

NAHR

Retrotransposons

V(D)J

SOURCE PRIORITY: the uploaded current faculty deck controls terminology, examples, and emphasis. The official Week 4 learning-objective sheet defines what must be mastered. This companion reorganizes the same material for retrieval and application without replacing the source deck.

Official Week 4 Objectives

1. Define recombination, explain its biological significance, and distinguish the four major recombination types.
2. Describe homologous recombination and non-homologous end joining (NHEJ), including their roles in double-strand-break repair, chromosome segregation, structural rearrangements, and antibody diversity.
3. Describe crossover frequency and relate genetic linkage to inheritance of HLA genes.
4. Explain how RNA transposons generate repetitive DNA and processed pseudogenes.
5. Explain how recombination contributes to immunoglobulin gene assembly and antibody diversity.

ASSIGNED SCHOLARRX

Chromosomal Translocations, Deletions & Inversions

Recombination is the physical exchange or rearrangement of DNA sequences. It is essential for DNA repair, restart of damaged replication forks, meiotic chromosome pairing/segregation, generation of genetic variation, integration of incoming DNA in bacteria, immune-receptor diversity, and genome evolution.

Type	Defining feature	High-yield role
Homologous recombination (HR)	Uses long stretches of homologous sequence	High-fidelity DSB repair; meiotic crossing over
Non-homologous end joining (NHEJ)	Directly rejoins broken ends with little/no homology	Major somatic-cell DSB repair; V(D)J joining; translocation risk
Transpositional recombination	Moves mobile DNA to new genomic sites	DNA transposons and retrotransposons; repetitive DNA
Conservative site-specific recombination	Cutting and rejoining at defined sequence sites	Research/engineering systems; precise rearrangement

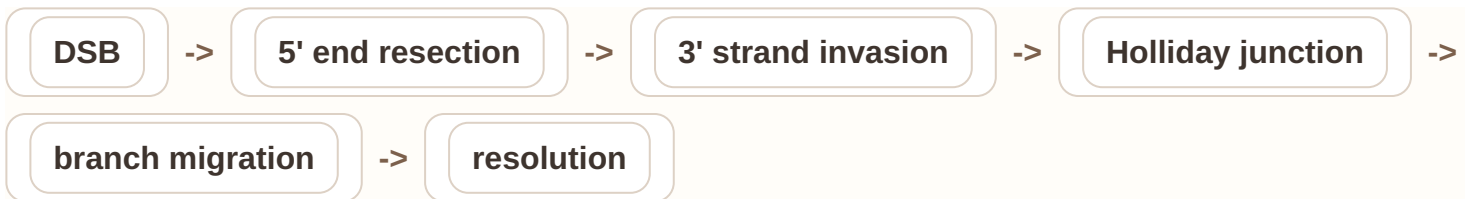
Big distinction: HR is sequence/homology driven and comparatively conservative. NHEJ is junction driven and can be imprecise. Transposition changes genomic position. Site-specific recombination requires defined target sequences.

When HR is favored

Requires relatively long homologous regions and is most important when a homologous template is available, especially in S/G2 and during meiosis.

Why meiosis needs it

Programmed double-strand breaks initiated by **Spo11** create recombination events that help homologous chromosomes pair and segregate correctly.



Consequences: crossover/reassortment, gene conversion, generation of new alleles, and accurate repair of broken DNA. In germ cells these changes can become heritable.

Do not over-memorize enzyme minutiae: the deck explicitly emphasizes the overall logic - break, resection, strand invasion/Holliday junction, migration, resolution.

Slides 12-13

Crossover frequency measures genetic distance

LO 3

Genes close together on the same chromosome recombine less often and therefore tend to be inherited together as a **haplotype**. Greater physical separation generally increases the chance of a crossover occurring between loci.

HLA anchor from the deck: HLA loci are physically close and usually inherited en bloc. The lecture uses the classic sibling-transplant relationship: a sibling has a 25% chance of sharing the same two parental HLA haplotypes; recombination within the HLA region is relatively uncommon but can break the expected haplotype.

Slides 14-16

When homologous recombination goes wrong

LO 2-3

Error	Mechanism	Clinical anchor in deck
Nondisjunction	Failure of homologs/sister chromatids to segregate properly	Aneuploidy such as trisomy 21 or monosomy X
Spo11 pathway defect	Failure to initiate normal meiotic recombination	Infertility/sterility

Non-allelic homologous recombination (NAHR)	Misaligned repetitive sequences recombine	Deletion, duplication, inversion, translocation
Unequal crossover	Homologous repeats pair out of register	PMP22 dosage disorders; factor VIII inversion example

Repetitive DNA creates recombination hot spots. Misalignment between similar repeats is a recurring mechanism for structural chromosome disease.

Slides 17-21

NHEJ: fast DSB repair with a genomic price

LO 2 & 5

Core mechanism

Broken ends are processed and directly ligated without a homologous template. Very short microhomologies can help alignment.

Cell-cycle context

Prominent in somatic cells and especially important in G1, when a sister chromatid is not available.

NHEJ is inherently error prone: trimming or filling DNA ends can create small insertions/deletions. Joining ends from different chromosomes can create oncogenic translocations.

HR	NHEJ
Template/homology driven	Template independent
High fidelity	More error prone
S/G2; meiosis especially important	Major somatic-cell pathway; especially G1
Misaligned homologous repeats can cause NAHR	Incorrectly paired DSBs can cause translocations

Cancer examples in the deck: t(9;22) BCR-ABL in chronic myeloid leukemia and t(8;14) involving MYC in Burkitt lymphoma illustrate how DSB misjoining can generate oncogenic fusion/regulatory events.

Immune connection: NHEJ repairs the DSBs created during antigen-receptor rearrangement. Defective NHEJ proteins can therefore impair V(D)J recombination and produce severe immunodeficiency phenotypes.

Slides 22-27

Transposons, retrotransposons, repetitive DNA, and processed pseudogenes

LO 4

Class	Mechanism	Copy-number effect	Major risk
DNA transposon	Cut-and-paste via transposase	Does not inherently increase copy number	DSB at donor site; insertional disruption
Retrotransposon	Copy-and-paste through RNA intermediate + reverse transcription	Increases copy number	Insertional mutation + expanded repetitive DNA/NAHR substrate

LINE-1

Long interspersed element; autonomous because it encodes proteins needed for its own reverse-transcription/integration pathway.

Alu / SINE

Short interspersed element; non-autonomous and borrows LINE-1 machinery. High copy number makes Alu sequences important substrates for ectopic recombination.

Processed pseudogene signature: reverse-transcribed cellular mRNA reinserted into DNA - therefore it typically lacks introns and its native promoter, may retain a 3' poly(A)-derived sequence, and can be flanked by target-site duplications.

Slide 28

Recombination creates antibody diversity

LO 5

RAG1/RAG2 recognize RSS

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DNA cleavage / DSB

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NHEJ rejoins coding ends

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junctional imprecision adds diversity

V(D)J recombination uses a cleavage chemistry related to ancient transposases. The intervening DNA is excised as a circle, while NHEJ joins selected antigen-receptor gene segments. Variation in segment choice plus imprecise end joining greatly expands receptor diversity.

Exam traps

Protect the mechanism-to-outcome links

All LOs

Trap 1: HR requires substantial homology; NHEJ does not.

Trap 2: HR is not synonymous with "crossing over" - it also performs template-directed DSB repair.

Trap 3: LINE-1 is autonomous; Alu/SINE is non-autonomous.

Trap 4: Processed pseudogenes come from RNA intermediates, so they usually lack introns.

Trap 5: V(D)J cleavage is RAG-mediated; joining is NHEJ-mediated.

30-SECOND FINISH

Four types: HR, NHEJ, transposition, site-specific. HR uses homology and Holliday-junction logic; NHEJ directly rejoins ends. Repeats predispose to NAHR. HLA genes are tightly linked. Retrotransposons expand repetitive DNA. RAG cleavage + NHEJ joining generates receptor diversity.